Relative degradation under Gaussian weight perturbation: muon vs adamw

chinchilla (token budget)
model size
muon - adamw degradation, % of own unperturbed loss
10
5
0
−5
−10

$\gamma = 0.003$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0138			
64	+0.0238				
32	+0.0138	+0.0100	+0.0114		
16	+0.0076	+0.0057	+0.0125		
8	+0.0055	+0.0045	+0.0074	+0.0015	
4	+0.0021	+0.0053	+0.0014	+0.0015	
2	+0.0022	+0.0033	+0.0016	+0.0025	+0.0018
1	+0.0022	+0.0020	+0.0017	+0.0012	+0.0004
0.5	+0.0012	+0.0009	-0.0004	+0.0008	+0.0003
0.25	-0.0004	+0.0013	+0.0013	-0.0008	+0.0006

$\gamma = 0.005$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0357			
64	+0.0626				
32	+0.0376	+0.0243	+0.0327		
16	+0.0229	+0.0150	+0.0304		
8	+0.0189	+0.0135	+0.0193	+0.0048	
4	+0.0098	+0.0155	+0.0070	+0.0034	
2	+0.0094	+0.0077	+0.0055	+0.0035	+0.0061
1	+0.0059	+0.0034	+0.0048	+0.0022	+0.0020
0.5	+0.0049	+0.0035	+0.0013	+0.0014	+0.0016
0.25	+0.0015	+0.0029	+0.0022	+0.0005	+0.0005

$\gamma = 0.007$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.0693			
64	+0.1151				
32	+0.0682	+0.0433	+0.0646		
16	+0.0417	+0.0297	+0.0569		
8	+0.0346	+0.0262	+0.0356	+0.0073	
4	+0.0150	+0.0281	+0.0127	+0.0029	
2	+0.0160	+0.0137	+0.0116	+0.0069	+0.0099
1	+0.0123	+0.0077	+0.0091	+0.0038	+0.0045
0.5	+0.0083	+0.0061	+0.0031	+0.0025	+0.0031
0.25	-0.0009	+0.0062	+0.0049	+0.0019	+0.0022

$\gamma = 0.01$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.1400			
64	+0.2132				
32	+0.1353	+0.0874	+0.1277		
16	+0.0848	+0.0623	+0.1083		
8	+0.0695	+0.0516	+0.0702	+0.0099	
4	+0.0326	+0.0567	+0.0250	+0.0092	
2	+0.0334	+0.0247	+0.0241	+0.0156	+0.0196
1	+0.0237	+0.0159	+0.0179	+0.0081	+0.0106
0.5	+0.0138	+0.0110	+0.0074	+0.0063	+0.0073
0.25	+0.0050	+0.0114	+0.0082	+0.0040	+0.0046

$\gamma = 0.02$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+0.5673			
64	+0.8906				
32	+0.5455	+0.3390	+0.5267		
16	+0.3492	+0.2508	+0.4255		
8	+0.2765	+0.2066	+0.2743	+0.0371	
4	+0.1350	+0.2199	+0.1039	+0.0424	
2	+0.1353	+0.0932	+0.0952	+0.0552	+0.0774
1	+0.0988	+0.0633	+0.0697	+0.0285	+0.0372
0.5	+0.0637	+0.0481	+0.0337	+0.0242	+0.0258
0.25	+0.0257	+0.0389	+0.0319	+0.0134	+0.0154

$\gamma = 0.03$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+1.3379			
64	+1.9916				
32	+1.2253	+0.7827	+1.2193		
16	+0.7845	+0.5756	+1.0067		
8	+0.6279	+0.4674	+0.6211	+0.0635	
4	+0.3097	+0.4993	+0.2388	+0.0950	
2	+0.3087	+0.2089	+0.2222	+0.1210	+0.1714
1	+0.2211	+0.1410	+0.1561	+0.0603	+0.0896
0.5	+0.1444	+0.1114	+0.0759	+0.0563	+0.0615
0.25	+0.0628	+0.0795	+0.0714	+0.0303	+0.0353

$\gamma = 0.04$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+2.5103			
64	+3.6321				
32	+2.1848	+1.4104	+2.2541		
16	+1.4341	+1.0568	+1.8468		
8	+1.1205	+0.8581	+1.1583	+0.1130	
4	+0.5650	+0.8985	+0.4336	+0.2047	
2	+0.5663	+0.3712	+0.4026	+0.2148	+0.3209
1	+0.3887	+0.2587	+0.2869	+0.1072	+0.1799
0.5	+0.2590	+0.1998	+0.1393	+0.0997	+0.1186
0.25	+0.1186	+0.1369	+0.1281	+0.0527	+0.0699

$\gamma = 0.05$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+4.1788			
64	+5.7946				
32	+3.5222	+2.2736	+3.6700		
16	+2.2685	+1.6931	+3.0059		
8	+1.7398	+1.3730	+1.8468	+0.1869	
4	+0.9122	+1.4275	+0.6903	+0.3260	
2	+0.8952	+0.5836	+0.6488	+0.3481	+0.5112
1	+0.6129	+0.4097	+0.4555	+0.1670	+0.2805
0.5	+0.4078	+0.3176	+0.2210	+0.1688	+0.1812
0.25	+0.1919	+0.2083	+0.1994	+0.0826	+0.1126

$\gamma = 0.07$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+9.4021			
64	+11.5420				
32	+7.0077	+4.7755	+8.0534		
16	+4.5048	+3.5777	+6.5387		
8	+3.4520	+2.8976	+3.9136	+0.3886	
4	+1.8783	+2.9412	+1.4495	+0.8043	
2	+1.7658	+1.1667	+1.3452	+0.7211	+1.1290
1	+1.2060	+0.8508	+0.9476	+0.3317	+0.6107
0.5	+0.7994	+0.6449	+0.4524	+0.3466	+0.3907
0.25	+0.3827	+0.4062	+0.4026	+0.1672	+0.2350

$\gamma = 0.1$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+22.0351			
64	+22.4779				
32	+14.7185	+10.8522	+19.8312		
16	+9.2147	+8.1515	+15.8965		
8	+7.1163	+6.6076	+9.1803	+0.9978	
4	+4.1096	+6.6664	+3.3971	+2.6375	
2	+3.7872	+2.5355	+3.0660	+1.6812	+2.8337
1	+2.4799	+1.9826	+2.1703	+0.7376	+1.5730
0.5	+1.6454	+1.4254	+0.9931	+0.7840	+0.9138
0.25	+0.8014	+0.8388	+0.8588	+0.3664	+0.5015

$\gamma = 0.13$

chinchilla (token budget)	30M	60M	100M	300M	600M
128		+36.3833			
64	+33.3247				
32	+24.4718	+20.1280	+39.5308		
16	+14.3700	+15.1949	+30.9760		
8	+11.9082	+12.1195	+18.3773	+1.7794	
4	+7.3662	+12.5315	+6.6057	+7.8893	
2	+6.7411	+4.5799	+5.9851	+3.3661	+6.0749
1	+4.2342	+3.8975	+4.2597	+1.4293	+3.4722
0.5	+2.8059	+2.6753	+1.8414	+1.5350	+1.8176
0.25	+1.4102	+1.4695	+1.5552	+0.6935	+0.9715

cell = 100 x (perturbed - unperturbed) / unperturbed for muon minus the same for adamw | negative (blue) = muon degrades less | white = no data

colour scale clipped at ±14.790 (97th pct); 12 cell(s) beyond it keep their printed value